

P PROJECT PART III

420-635-AB-Network Installation and Administration I

OBJECTIVES

In this 3rd part of the project, you will **optimize the performance of an Apache server, view usage statistics, create scripts with cgi and php, and configure a secure server.**

GUIDELINES

The **Project Part 3** is graded out of **100** and accounts for **11%** of the final score.

You must provide a **detailed report** documenting the **commands executed in each section, the results obtained, and any relevant remarks. (Include clear and prominent screenshots.)**

You must also submit a **compressed file** containing of the **/etc/httpd/conf/httpd.conf** file and containing of **/var/www/html_project3** and **/var/www/cgi-bin** directories.

The report must be submitted upon completion and before the deadline. Submission must be done via **Moodle**. The report file must be named: **FirstName_LastName_Report_P3.pdf**.

PONDERATION

Task	Graded on
Task 1 – Multi-Process Module (MPM)	15%
Task 2 – Apache Server Monitoring	10%
Task 3 – CGI	15%
Task 4 – PHP	15%
Task 5 – MYSQL/PHP	20%
Task 6 – SSL	10%
Report quality	15%
Total	100%

TASK 1 – MULTI-PROCESS MODULE (MPM) (15 POINTS)

- The web pages for this project should be in the directory:
/var/www/html_project3
- Add links to all the web pages of this project in the following file:
var/www/html_project3/master_project3.html. *See the example provided on the last page of this document.*

- Configure your Apache web server with the following parameters:
 - Start with 12 httpd server processes when the service starts.
 - Maintain a minimum of 6 idle server processes at all times.
 - Allow a maximum of 12 idle server processes.
 - Handle up to 180 simultaneous client requests.
 - Queue a maximum of 100 pending requests when the maximum of 180 simultaneous requests is reached.
 - Enable persistent (KeepAlive) connections, with:
 - A maximum of 50 consecutive requests per connection.
 - A 20-second timeout between two consecutive requests from the same client.
 - Limit the maximum request duration to 55 seconds per client request.

TASK 2 – APACHE SERVER MONITORING (10 POINTS)

- Enable **ExtendedStatus** to allow detailed monitoring of Apache server activity.
- Configure the **server-status** and **server-info** handlers so that they are accessible only to users on the **192.168.50.0/24** subnet.

TASK 3 – CGI - (15 POINTS)

- Configure your web server to use **cgi-bins**.
- Create a **Perl script** that displays the text **"This is Task 3 (CGI)"** in the web browser.
- Place this script in the **/var/www/cgi-bin** directory.

TASK 4 – PHP - (15 POINTS)

- Configure your web server to use **PHP**.
- Create a **PHP script** that displays the message **"Hello my friend!"** in the **web browser** for users whose IP address belongs to the **192.168.100.0/24** subnet and displays **"Hello stranger!"** for all other users outside this subnet.
- Place this PHP script in the **/var/www/html_project3/q4** directory.

TASK 5 – MYSQL/PHP (20 POINTS)

1. Install **MySQL** and create a database **company**.
2. Create in this database an “**employees**” table that contains two fields: **name** and **salary**.
3. Insert several records into the table, then verify the contents.
4. Create a **PHP script** that connects to this database and displays the contents of the **employees** table in the web browser as an **HTML table**.
5. Place this PHP script in the **/var/www/html_project3/q5** directory.

TASK 6 – SSL (10 POINTS)

1. Configure your web server to use **SSL**.
2. Test access to your server using the **https** security protocol by navigating to:
https://serverX (where **X** is your assigned server number).

master_project3.html

Project Part 3

Task 2

[To view the server status](#)

[To view the server information](#)

Task 3

[To execute the CGI Script](#)

Task 4

[To execute the PHP script using 192.168.100.1](#)

[To execute the PHP script using another subnet](#)

Task 5

[To list the employees mysql table](#)

Task 6

[To view the website in SSL](#)